

7.1 | BASIC STRUCTURE OF THE pn JUNCTION

Figure 7.1a schematically shows the pn junction. It is important to realize that the entire semiconductor is a single-crystal material in which one region is doped with acceptor impurity atoms to form the p region and the adjacent region is doped with donor atoms to form the n region. The interface separating the n and p regions is referred to as the metallurgical junction.

The impurity doping concentrations in the p and n regions are shown in Figure 7.1b. For simplicity, we will consider a step junction in which the doping concentration is uniform in each region and there is an abrupt change in doping at the junction. Initially, at the metallurgical junction, there is a very large density gradient in both electron and hole concentrations. Majority carrier electrons in the n region will begin diffusing into the p region, and majority carrier holes in the p region will begin diffusing into the n region. If we assume there are no external connections to the semiconductor, then this diffusion process cannot continue indefinitely. As electrons diffuse from the n region, positively charged donor atoms are left behind. Similarly, as holes diffuse from the p region, they uncover negatively charged acceptor atoms. The net positive and negative charges in the n and p regions induce an electric field in the region near the metallurgical junction, in the direction from the positive to the negative charge, or from the n to the p region.

The net positively and negatively charged regions are shown in Figure 7.2. These two regions are referred to as the space charge region. Essentially all electrons and holes are swept out of the space charge region by the electric field. Since the space charge region is depleted of any mobile charge, this region is also referred to as the

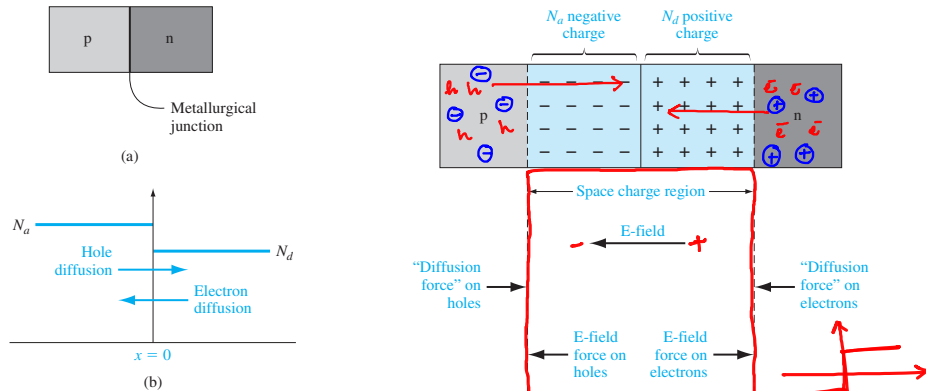


Figure 7.1 | (a) Simplified geometry of a pn junction; (b) doping profile of an ideal uniformly doped pn junction.

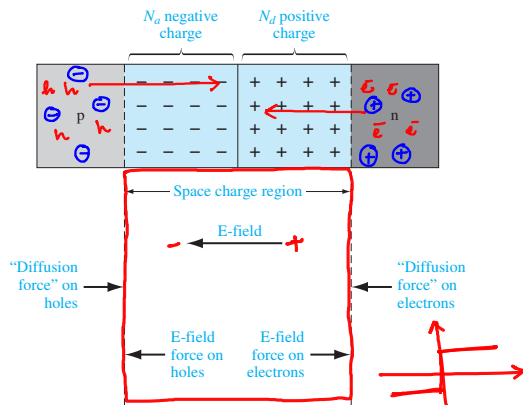


Figure 7.2 | The space charge region, the electric field, and the forces acting on the charged carriers.

depletion region; these two terms will be used interchangeably. Density gradients still exist in the majority carrier concentrations at each edge of the space charge region. We can think of a density gradient as producing a diffusion force that acts on the majority carriers. These diffusion forces, acting on the electrons and holes at the edges of the space charge region, are shown in the figure. The electric field in the space charge region produces another force on the electrons and holes, which is in the opposite direction to the diffusion force for each type of particle. In thermal equilibrium, the diffusion force and the E-field force exactly balance each other.

Under Thermal Equilibrium;
Mathematically: $F_{diff} = E$ due to space charge region !!

7.2 | ZERO APPLIED BIAS

We have considered the basic pn junction structure and discussed briefly how the space charge region is formed. In this section we will examine the properties of the step junction in thermal equilibrium, where no currents exist and no external excitation is applied. We will determine the space charge region width, electric field, and potential through the depletion region.

The analysis in this chapter is based on two assumptions that we have considered in previous chapters. The first assumption is that the Boltzmann approximation is valid, which means that each semiconductor region is nondegenerately doped. The second assumption is that complete ionization exists, which means that the temperature of the pn junction is not "too low."

Means; there is some energy !! Because

$$E_g \approx \frac{k_B T}{2e}$$

7.2.1 Built-in Potential Barrier

If we assume that no voltage is applied across the pn junction, then the junction is in thermal equilibrium—the Fermi energy level is constant throughout the entire system. Figure 7.3 shows the energy-band diagram for the pn junction in thermal equilibrium. The conduction and valence band energies must bend as we go through the space charge region, since the relative position of the conduction and valence bands with respect to the Fermi energy changes between p and n regions.

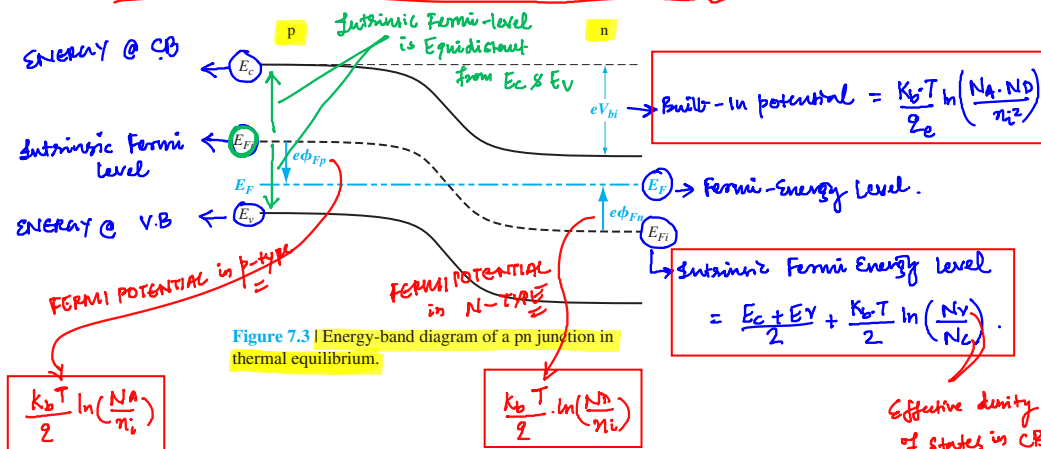


Figure 7.3 | Energy-band diagram of a pn junction in thermal equilibrium.

Effective density of states in CB = VB

① CONCEPT OF FERMI-ENERGY LEVEL (E_F):

Fermi-Energy level is the ENERGY LEVEL at which the probability of finding an e^- is maximum; (60)
Atleast 50% @ Absolute zero temperature (0K).

Because $E_g \approx \frac{K_b T}{2e}$; Hence; if $T \uparrow$, $N_e \uparrow$; So if with "NO ANY THERMAL ENERGY"; e^- 's probability
Carrier excitation is there; that means our supply energy would be
less Required that is later on will be good for

In INTRINSIC S/C $\rightarrow E_F$ lies in the middle of VB & CB. the device. V_T will be low...

In N-TYPE $\rightarrow$ Near the CB, i.e Donor level ionization.

In P-TYPE $\rightarrow$ Near the VB i.e Acceptor level ionization.

It is represented by $\rightarrow$

$$f(E) = \text{probability of Availability of } e^-s = \frac{1}{1 + e^{(E-E_F)/k_b T}}$$

Electrons in the conduction band of the n region see a potential barrier in trying to move into the conduction band of the p region. This potential barrier is referred to as the built-in potential barrier and is denoted by V_{bi} . The built-in potential barrier maintains equilibrium between majority carrier electrons in the n region and minority carrier electrons in the p region, and also between majority carrier holes in the p region and minority carrier holes in the n region. This potential difference across the junction cannot be measured with a voltmeter because new potential barriers will be formed between the probes and the semiconductor that will cancel V_{bi} . The potential V_{bi} maintains equilibrium, so no current is produced by this voltage.

② The intrinsic Fermi level is equidistant from the conduction band edge through the junction; thus, the built-in potential barrier can be determined as the difference between the intrinsic Fermi levels in the p and n regions. We can define the potentials ϕ_{Fn} and ϕ_{Fp} as shown in Figure 7.3, so we have

$$V_{bi} = |\phi_{Fn}| + |\phi_{Fp}| \quad (7.1)$$

In the n region, the electron concentration in the conduction band is given by

Effective density of state in C.B. $\leftarrow$

$$n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right] \quad (7.2)$$

which can also be written in the form

$$n_0 = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right] \quad (7.3)$$

In terms of n_i : intrinsic Carrier Conc.

where n_i and E_{Fi} are the intrinsic carrier concentration and the intrinsic Fermi energy, respectively. We may define the potential ϕ_{Fn} in the n region as

$$e\phi_{Fn} = E_{Fi} - E_F \quad (7.4)$$

Equation (7.3) may then be written as

$$n_0 = n_i \exp\left[\frac{-(e\phi_{Fn})}{kT}\right] \quad (7.5)$$

Taking the natural log of both sides of Equation (7.5), setting $n_0 = N_d$ and solving for the potential, we obtain

$$\phi_{Fn} = \frac{-kT}{e} \ln\left(\frac{N_d}{n_i}\right) \quad (7.6)$$

Because n_i is the e^- conc. in N-side.

Similarly, in the p region, the hole concentration is given by

$$p_0 = N_a = n_i \exp\left[\frac{E_{Fi} - E_F}{kT}\right] \quad (7.7)$$

where N_a is the acceptor concentration. We can define the potential ϕ_{Fp} in the p region as

$$e\phi_{Fp} = E_{Fi} - E_F \quad (7.8)$$

Combining Equations (7.7) and (7.8), we find that

$$\phi_{rp} = +\frac{kT}{e} \ln \left(\frac{N_a}{n_i} \right) \quad (7.9)$$

Finally, the built-in potential barrier for the step junction is found by substituting Equations (7.6) and (7.9) into Equation (7.1), which yields

$$V_{bi} = \frac{kT}{e} \ln \left(\frac{N_a N_d}{n_i^2} \right) = V_i \ln \left(\frac{N_a N_d}{n_i^2} \right) \quad (7.10)$$

where $V_i = kT/e$ and is defined as the **thermal voltage**.

At this time, we should note a subtle but important point concerning notation. Previously in the discussion of a semiconductor material, N_d and N_a denoted donor and acceptor impurity concentrations in the same region, thereby forming a compensated semiconductor. From this point on in the text, N_d and N_a will denote the net donor and acceptor concentrations in the individual n and p regions, respectively. If the p region, for example, is a compensated material, then N_a will represent the difference between the actual acceptor and donor impurity concentrations. (The parameter N_d is defined in a similar manner for the n region.)

NET

Objective: Calculate the built-in potential barrier in a pn junction.

Consider a silicon pn junction at $T = 300$ K with doping concentrations of $N_a = 2 \times 10^{17} \text{ cm}^{-3}$ and $N_d = 10^{15} \text{ cm}^{-3}$.

■ Solution

The built-in potential barrier is determined from Equation (7.10) as

$$V_{bi} = V_i \ln \left(\frac{N_a N_d}{n_i^2} \right) = (0.0259) \ln \left[\frac{(2 \times 10^{17})(10^{15})}{(1.5 \times 10^{10})^2} \right] = 0.713 \text{ V}$$

If we change the doping concentration in the p region of the pn junction such that the doping concentrations become $N_a = 10^{16} \text{ cm}^{-3}$ and $N_d = 10^{15} \text{ cm}^{-3}$, then the built-in potential barrier becomes $V_{bi} = 0.635 \text{ V}$.

■ Comment

The built-in potential barrier changes only slightly as the doping concentrations change by orders of magnitude because of the logarithmic dependence.

■ EXERCISE PROBLEM

Ex 7.1 (a) Calculate the built-in potential barrier in a silicon pn junction at $T = 300$ K for (i) $N_a = 5 \times 10^{15} \text{ cm}^{-3}$, $N_d = 10^{17} \text{ cm}^{-3}$ and (ii) $N_a = 2 \times 10^{16} \text{ cm}^{-3}$, $N_d = 2 \times 10^{15} \text{ cm}^{-3}$. (b) Repeat part (a) for a GaAs pn junction.

[LA 111 (!) 'A 071 (!) (q) :A 1L9'0 (!!) 'A 9EL'0 (!) (v) 'suV]

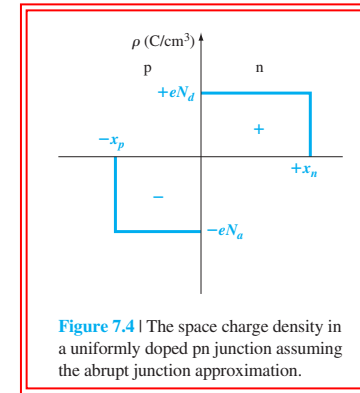


Figure 7.4 | The space charge density in a uniformly doped pn junction assuming the abrupt junction approximation.

7.2.2 Electric Field

An electric field is created in the depletion region by the separation of positive and negative space charge densities. Figure 7.4 shows the volume charge density distribution in the pn junction assuming uniform doping and assuming an abrupt junction approximation. We will assume that the space charge region abruptly ends in the n region at $x = +x_n$ and abruptly ends in the p region at $x = -x_p$ (x_p is a positive quantity).

The electric field is determined from Poisson's equation, which, for a one-dimensional analysis, is

$$\frac{d^2 \phi(x)}{dx^2} = \frac{-\rho(x)}{\epsilon_s} = -\frac{dE(x)}{dx} \quad (7.11)$$

where $\phi(x)$ is the electric potential, $E(x)$ is the electric field, $\rho(x)$ is the volume charge density, and ϵ_s is the permittivity of the semiconductor. From Figure 7.4, the charge densities are

$$\rho(x) = -eN_a \quad -x_p < x < 0 \quad (7.12a)$$

and

$$\rho(x) = eN_d \quad 0 < x < x_n \quad (7.12b)$$

The electric field in the p region is found by integrating Equation (7.11). We have

$$E = \int \frac{\rho(x)}{\epsilon_s} dx = - \int \frac{eN_a}{\epsilon_s} dx = \frac{-eN_a}{\epsilon_s} x + C_1 \quad (7.13)$$

where C_1 is a constant of integration. The electric field is assumed to be zero in the neutral p region for $x < -x_p$ since the currents are zero in thermal equilibrium. Since there are no surface charge densities within the pn junction structure, the electric

field is a continuous function. The constant of integration is determined by setting $E = 0$ at $x = -x_p$. The electric field in the p region is then given by

$$E = -\frac{eN_d}{\epsilon_s}(x + x_p) \quad -x_p \leq x \leq 0 \quad (7.14)$$

In the n region, the electric field is determined from

$$E = \int \frac{(eN_d)}{\epsilon_s} dx = \frac{eN_d}{\epsilon_s}x + C_2 \quad (7.15)$$

where C_2 is again a constant of integration and is determined by setting $E = 0$ at $x = x_n$, since the E-field is assumed to be zero in the n region and is a continuous function. Then

$$E = -\frac{eN_d}{\epsilon_s}(x_n - x) \quad 0 \leq x \leq x_n \quad (7.16)$$

The electric field is also continuous at the metallurgical junction, or at $x = 0$. Setting Equations (7.14) and (7.16) equal to each other at $x = 0$ gives

$$N_ax_p = N_dx_n \quad (7.17)$$

Equation (7.17) states that the number of negative charges per unit area in the p region is equal to the number of positive charges per unit area in the n region.

Figure 7.5 is a plot of the electric field in the depletion region. The electric field direction is from the n to the p region, or in the negative x direction for this geometry. For the uniformly doped pn junction, the E-field is a linear function of distance through the junction, and the maximum (magnitude) electric field occurs at the metallurgical junction. An electric field exists in the depletion region even when no voltage is applied between the p and n regions.

The potential in the junction is found by integrating the electric field. In the p region then, we have

$$\phi(x) = -\int E(x)dx = \int \frac{eN_d}{\epsilon_s}(x + x_p)dx \quad (7.18)$$

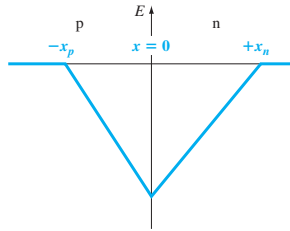


Figure 7.5 | Electric field in the space charge region of a uniformly doped pn junction.

or

$$\phi(x) = \frac{eN_d}{\epsilon_s} \left(\frac{x^2}{2} + x_p \cdot x \right) + C'_1 \quad (7.19)$$

where C'_1 is again a constant of integration. The potential difference through the pn junction is the important parameter, rather than the absolute potential, so we may arbitrarily set the potential equal to zero at $x = -x_p$. The constant of integration is then found as

$$C'_1 = \frac{eN_d}{2\epsilon_s}x_p^2 \quad (7.20)$$

so that the potential in the p region can now be written as

$$\phi(x) = \frac{eN_d}{2\epsilon_s}(x + x_p)^2 \quad (-x_p \leq x \leq 0) \quad (7.21)$$

The potential in the n region is determined by integrating the electric field in the n region, or

$$\phi(x) = \int \frac{eN_d}{\epsilon_s}(x_n - x)dx \quad (7.22)$$

Then

$$\phi(x) = \frac{eN_d}{\epsilon_s} \left(x_n \cdot x - \frac{x^2}{2} \right) + C'_2 \quad (7.23)$$

where C'_2 is another constant of integration. The potential is a continuous function, so setting Equation (7.21) equal to Equation (7.23) at the metallurgical junction, or at $x = 0$, gives

$$C'_2 = \frac{eN_d}{2\epsilon_s}x_p^2 \quad (7.24)$$

The potential in the n region can thus be written as

$$\phi(x) = \frac{eN_d}{\epsilon_s} \left(x_n \cdot x - \frac{x^2}{2} \right) + \frac{eN_d}{2\epsilon_s}x_p^2 \quad (0 \leq x \leq x_n) \quad (7.25)$$

Figure 7.6 is a plot of the potential through the junction and shows the quadratic dependence on distance. The magnitude of the potential at $x = x_n$ is equal to the built-in potential barrier. Then from Equation (7.25), we have

$$V_{bi} = |\phi(x = x_n)| = \frac{e}{2\epsilon_s} (N_dx_n^2 + N_ax_p^2) \quad (7.26)$$

The potential energy of an electron is given by $E = -e\phi$, which means that the electron potential energy also varies as a quadratic function of distance through the space charge region. The quadratic dependence on distance was shown in the energy-band diagram of Figure 7.3, although we did not explicitly know the shape of the curve at that time.

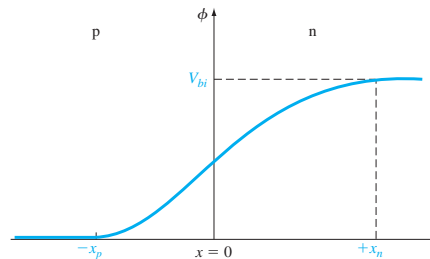


Figure 7.6 | Electric potential through the space charge region of a uniformly doped pn junction.

7.2.3 Space Charge Width

We can determine the distance that the space charge region extends into the p and n regions from the metallurgical junction. This distance is known as the space charge width. From Equation (7.17), we may write, for example,

$$x_p = \frac{N_d x_n}{N_a} \quad (7.27)$$

Then, substituting Equation (7.27) into Equation (7.26) and solving for x_n , we obtain

$$x_n = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[\frac{N_a}{N_d} \right] \left[\frac{1}{N_a + N_d} \right] \right\}^{1/2} \quad (7.28)$$

Equation (7.28) gives the space charge width, or the width of the depletion region, x_n extending into the n-type region for the case of zero applied voltage.

Similarly, if we solve for x_n from Equation (7.17) and substitute into Equation (7.26), we find

$$x_p = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[\frac{N_d}{N_a} \right] \left[\frac{1}{N_a + N_d} \right] \right\}^{1/2} \quad (7.29)$$

where x_p is the width of the depletion region extending into the p region for the case of zero applied voltage.

The total depletion or space charge width W is the sum of the two components, or

$$W = x_n + x_p \quad (7.30)$$

Using Equations (7.28) and (7.29), we obtain

$$W = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[\frac{N_a + N_d}{N_a N_d} \right] \right\}^{1/2} \quad (7.31)$$

The built-in potential barrier can be determined from Equation (7.10), and then the total space charge region width is obtained using Equation (7.31).

EXAMPLE 7.2

Objective: Calculate the space charge width and electric field in a pn junction for zero bias.

Consider a silicon pn junction at $T = 300$ K with doping concentrations of $N_a = 10^{16} \text{ cm}^{-3}$ and $N_d = 10^{15} \text{ cm}^{-3}$.

■ Solution

In Example 7.1, we determined the built-in potential barrier as $V_{bi} = 0.635$ V. From Equation (7.31), the space charge width is

$$\begin{aligned} W &= \left[\frac{2\epsilon_i V_{bi}}{e} \left[\frac{N_a + N_d}{N_a N_d} \right] \right]^{1/2} \\ &= \left[\frac{2(11.7)(8.85 \times 10^{-14})(0.635)}{1.6 \times 10^{-19}} \left[\frac{10^{16} + 10^{15}}{(10^{16})(10^{15})} \right] \right]^{1/2} \\ &= 0.951 \times 10^{-4} \text{ cm} = 0.951 \text{ } \mu\text{m} \end{aligned}$$

Using Equations (7.28) and (7.29), we can find $x_n = 0.8644 \mu\text{m}$, and $x_p = 0.0864 \mu\text{m}$.

The peak electric field at the metallurgical junction, using Equation (7.16) for example, is

$$E_{\max} = -\frac{eN_d x_n}{\epsilon_s} = -\frac{(1.6 \times 10^{-19})(10^{15})(0.8644 \times 10^{-4})}{(11.7)(8.85 \times 10^{-14})} = -1.34 \times 10^4 \text{ V/cm}$$

■ Comment

The peak electric field in the space charge region of a pn junction is quite large. We must keep in mind, however, that there is no mobile charge in this region; hence there will be no drift current. We may also note, from this example, that the width of each space charge region is a reciprocal function of the doping concentration: The depletion region will extend further into the lower-doped region.

■ EXERCISE PROBLEM

Ex 7.2 A silicon pn junction at $T = 300$ K with zero applied bias has doping concentrations of $N_d = 5 \times 10^{16} \text{ cm}^{-3}$ and $N_a = 5 \times 10^{15} \text{ cm}^{-3}$. Determine x_n , x_p , W , and $|E_{\max}|$.

$$\text{(Ans. } x_n = 4.11 \times 10^{-6} \text{ cm, } x_p = 4.11 \times 10^{-5} \text{ cm, } W = 4.52 \times 10^{-5} \text{ cm, } |E_{\max}| = 3.18 \times 10^4 \text{ V/cm)}$$

TEST YOUR UNDERSTANDING

TYU 7.1 Calculate V_{bi} , x_n , x_p , W , and $|E_{\max}|$ for a silicon pn junction at zero bias and $T = 300$ K for doping concentrations of (a) $N_a = 2 \times 10^{17} \text{ cm}^{-3}$, $N_d = 10^{16} \text{ cm}^{-3}$ and

(b) $N_a = 4 \times 10^{15} \text{ cm}^{-3}$, $N_d = 3 \times 10^{16} \text{ cm}^{-3}$.

$$[\text{AHS}, (a) V^{bi} = 0.772 \text{ V}, x_n^{bi} = 0.3085 \mu\text{m}, x_p^{bi} = 0.0154 \mu\text{m}, W = 0.3240 \mu\text{m}, E_{\text{max}}^{bi} = 4.77 \times 10^4 \text{ V/cm}, (b) V^{bi} = 0.699 \text{ V}, x_n^{bi} = 0.0596 \mu\text{m}, x_p^{bi} = 0.4469 \mu\text{m}, W = 0.5064 \mu\text{m}, E_{\text{max}}^{bi} = 2.76 \times 10^4 \text{ V/cm}]$$

TYU 7.2 Repeat Exercise Problem Ex 7.2 for a GaAs pn junction. $|E_{\text{max}}| = 3.86 \times 10^5 \text{ V/cm}$, $V_{bi} = 1.186 \text{ V}$, $x_n = 0.05590 \mu\text{m}$, $x_p = 0.5590 \mu\text{m}$, $W = 0.6149 \mu\text{m}$.